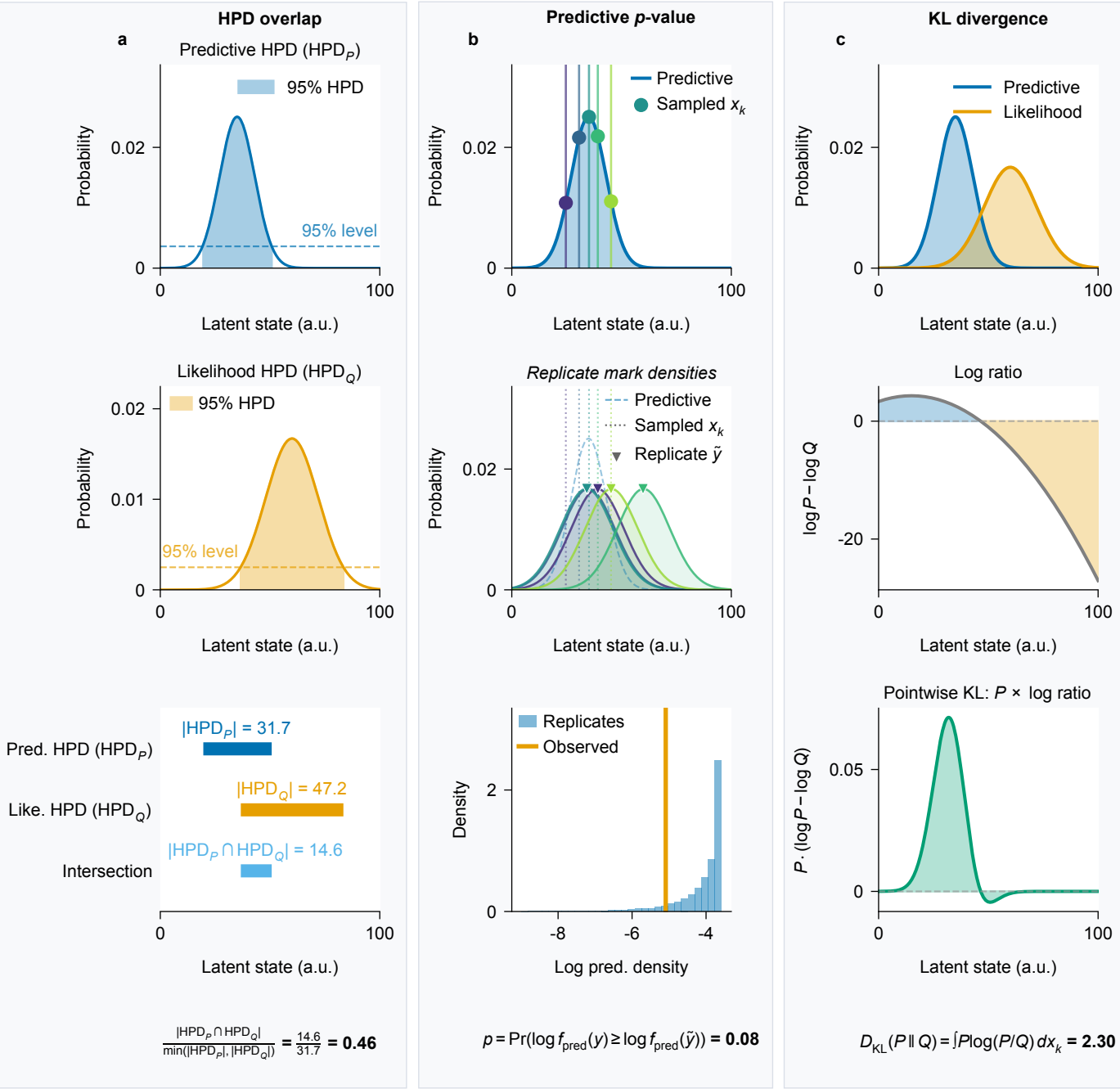

HPD overlap
a
Predictive HPD (HPD$_P$)
95% HPD
95% level
Probability
0.02
0
0
100
Latent state (a.u.)
Likelihood HPD (HPD$_Q$)
95% HPD
0.02
0.01
95% level
Probability
0
0
100
Latent state (a.u.)
Pred. HPD (HPD$_P$)
Like. HPD (HPD$_Q$)
Intersection
|HPD$_P$| = 31.7
|HPD$_Q$| = 47.2
|HPD$_P$ ∩ HPD$_Q$| = 14.6
0
100
Latent state (a.u.)
$\frac{|\text{HPD}_P \cap \text{HPD}_Q|}{\min(|\text{HPD}_P|, |\text{HPD}_Q|)} = \frac{14.6}{31.7} = \mathbf{0.46}$
Predictive p-value
b
Predictive
Sampled x_k
Probability
0.02
0
0
100
Latent state (a.u.)
Replicate mark densities
Predictive
Sampled x_k
Replicate $\tilde{y}$
Probability
0.02
0
0
100
Latent state (a.u.)
Replicates
Observed
Density
2
0
-8
-6
-4
Log pred. density
$p = \Pr(\log f_{\text{pred}}(y) \geq \log f_{\text{pred}}(\tilde{y})) = \mathbf{0.08}$
KL divergence
c
Predictive
Likelihood
Probability
0.02
0
0
100
Latent state (a.u.)
Log ratio
0
-20
$\log P - \log Q$
0
100
Latent state (a.u.)
Pointwise KL: P × log ratio
$P \cdot (\log P - \log Q)$
0.05
0
0
100
Latent state (a.u.)
$D_{\text{KL}}(P \| Q) = \int P \log(P/Q)\, dx_k = \mathbf{2.30}$